(Please write your Enrollment Number)

MID-TERM EXAMINATION
(Course Name B. Tech., ECE) (Semester: 3rd)
(October, 2025) OFF LINE mode

Subject Code: BEC 202	Subject: Analog Communication Systems
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)
	(a) If the probability of a bad reaction from medicine is 0.002, determine the chance that out of 1000 person more than 3 will suffer a bad reaction from medicine, using Poisson Distribution.	
	b) Define mean or average or expected value of a random variable. Explain how it is calculated for continuous and discrete random variables.	
	c) Determine the percentage power saving when the carrier wave and one of the side bands are suppressed in an AM wave modulated to a depth of (i) 100% (ii) 50%.	
	d) Describe VSB transmission. Give its applications.	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) Explain the properties of Fourier analysis of continuous and discrete signal.	
	b) Explain Uniform distribution and Gaussian distribution.	
	c) Write short note on (i) Thermal noise (ii) White noise.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Explain the working of Phase shift method of SSB with the help of block diagram. List its advantages w.r.t. the other methods.	
	b) Sketch and explain the working of QAM or Quadrature Carrier Multiplexing.	
	c) With the help of block diagram explain the Envelop Detector method for AM generation.	